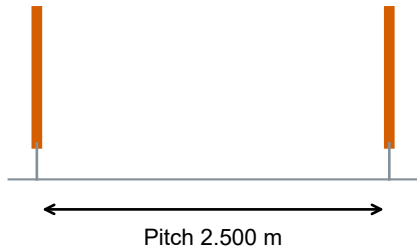
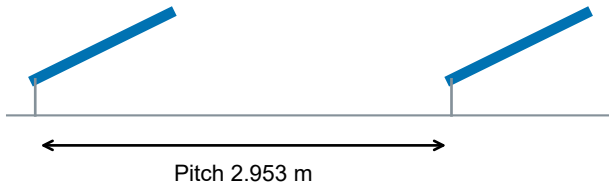


Same lower edge: 0.30 m | Same clear aisle: 2.00 m

Vertical bifacial



Tilted monofacial



Shared geometry + TMY

No weeds / snow

pvlib hourly AC

Clean energy + height sweep

ESS / cost screening

Equipment aging; clean PV

Warp GPU

Dry + rain first-hit tracing

Persistent dust pools

Assumed rain wash-off

10-year optical losses

HSU domain checked

Annual clean yield, assumed soiling loss and financial screening have distinct outputs; no field validation.